

Sara Winsemius

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Education

University of California, Davis

PhD in Ecology, March 2025

Advisors: Dr. Yufang Jin and Dr. Hugh Safford

Thesis: *A 10,000' view for detailed assessment: Remote sensing reveals patterns of forest structure and fire severity*

University of California, Berkeley

Bachelor of Science: Environmental Sciences, May 2013

Minor: Forestry and Natural Resources

Thesis: *Plant-soil interactions and implications for restoration of coastal sand dunes at Point Reyes National Seashore.*

Publications

Winsemius, S., Y. Jin, H. D. Safford, M. C. Agne, R. A. Andrus, A. C. Cansler, A. C. Caprio, W. W. Covington, C. Farris, P. Z. Fulé, B. J. Harvey, S. M. Hood, D. W. Huffman, E. J. McClure, J. P. Roccaforte, S. J. Saber, M. T. Stoddard, L. Trader, P. J. van Mantgem, M. C. Wright, and M. J. Koontz. Basal area loss from fire using field-calibrated remote sensing refines western U.S. fire severity measurements. In review.

Winsemius, S., C. Babcock, V. R. Kane, K. J. Bormann, H. D. Safford, and Y. Jin. 2024. Improved aboveground biomass estimation and regional assessment with aerial lidar in California's subalpine forests. *Carbon Balance and Management* 19:41.

Winsemius, S., and J. Braaten. "Zonal Statistics." In *Cloud-Based Remote Sensing with Google Earth Engine: Fundamentals and Applications*, edited by Jeffrey A. Cardille, Morgan A. Crowley, David Saah, and Nicholas E. Clinton, 1st ed. Springer, 2023.

Brodie, E. G., J. A. E. Stewart, **S. Winsemius**, J. E. D. Miller, A. M. Latimer, and H. D. Safford. Wildfire Facilitates Upslope Advance in a Shade-Intolerant but Not a Shade-Tolerant Conifer. *Ecological Applications* 33, no. 5 (2023): e2888. <https://doi.org/10.1002/eap.2888>.

Ward, L. W., M. L. Hladik, A. Guzman, **S. Winsemius**, A. Bautista, C. Kremen, and N. J. Mills. 2022. Pesticide exposure of wild bees and honey bees foraging from field border flowers in intensively managed agriculture areas. *Science of the Total Environment*. doi: 10.1016/j.scitotenv.2022.154697.

Posters

Winsemius S, Jin Y, Safford H, Koontz K. Basal area loss from wildfire using field-calibrated remote sensing refines western U.S. fire severity measurements. Association for Fire Ecology Congress, December 2023.

- Winsemius S**, Safford H, Jin Y. Assessing disturbance severity with lags in post-fire mortality in subalpine forests of the Sierra Nevada. American Geophysical Union Conference, December 2020.
- Winsemius S**, Jin Y, Safford H. Threats for carbon storage in high elevation forests of the Sierra Nevada. American Geophysical Union Conference, December 2019.
- Winsemius S**, Jin Y, Safford H. Threats for carbon storage in high elevation forests of the Sierra Nevada. Sierra Nevada Climate Change Symposium, September 2019.
- Winsemius S**, Jin Y, Safford H. The role of fire in shaping subalpine forest structure and carbon dynamics. Sequoia Science Symposium, November 2018.
Also presented at the Graduate Student Symposium in Ecology, February 2019.
- Winsemius S***, Ward L*, Hladik M, Sardiñas H, Mills N, Kremen C. Exploring potential routes of neonicotinoid exposure within pollinator hedgerows adjacent to seed-treated sunflower. Keeping Bees Healthy Symposium, May 2016. *authors contributed equally
- Winsemius S**, Stein C, Parsons L, Ryan A, Suding K. Plant-soil interactions and implications for restoration of coastal sand dunes at Point Reyes National Seashore. California Native Plant Society Conference, January 2015.

Oral Presentations

- Winsemius S**. Fire severity mapping with a new open source tool. California Forest Science Symposium. March 2025.
- Winsemius S**. Fire, carbon, and climate change in California's high elevation forests. Natural Areas Conference, Sierra to Sagebrush: Integrating Management and Stewardship Across Landscapes. October 2020.
- Winsemius S**. Invasive plant-soil interactions and implications for dune restoration and native species at Point Reyes National Seashore. Red Barn Talk Lecture Series. Point Reyes National Seashore, November 2013.
- Winsemius S**. Strategies for restoration of coastal sand dunes at Point Reyes National Seashore." Environmental Science Senior Thesis Symposium. UC Berkeley, May 2013.

Grants and Awards

2020. Sequoia Parks Conservancy. Sequoia Science Learning Center Research Grants Program.
"Threats for carbon storage in high montane forests: evaluating key drivers of change in the Sierra Nevada." \$2,400
2019. CAL FIRE Forest Health Grant Program. "Threats for carbon storage in high montane forests in the Sierra Nevada." \$66,892
2019. Henry A. Jastro Research Fellowship. UC Davis College of Ag & Environmental Sciences.
"Threats for carbon storage in high montane forests: evaluating key drivers and mapping change in the Sierra Nevada." \$3,000

2019. Sequoia Parks Conservancy. Sequoia Science Learning Center Research Grants Program.
 "Threats for carbon storage in high montane forests: evaluating key drivers of change in the Sierra Nevada." \$4,500

2018. Sequoia Parks Conservancy. Sequoia Science Learning Center Research Grants Program.
 "Changing Disturbances Regimes in Subalpine Forests of the Sierra Nevada." \$3,000

2016. First prize in graduate student poster session, Keeping Bees Healthy Symposium. \$1,000

2009-2013. UC Berkeley Regents' and Chancellor's Scholarship.

Research Experience

ORISE, USFS Rocky Mountain Research Station <i>Postdoctoral Research Fellow</i>	April 2025 – present
UC Davis Jin Lab <i>Graduate Student Researcher</i>	January 2018 – March 2025
UC Berkeley Kremen Lab <i>Staff Research Associate/Lab Manager</i>	February 2015 – September 2017
UC Berkeley Kremen Lab <i>Research Assistant</i>	June – October 2014
Point Reyes National Seashore <i>Biological Technician</i>	March – May 2014
Point Reyes National Seashore Association <i>Dune Restoration Intern</i>	May – August 2012 June – November 2013
UC Berkeley Resh Lab <i>Research Lab Assistant</i>	September 2011 – May 2012
Lava Beds National Monument <i>Student Conservation Association, Landscape Restoration Intern</i>	May – August 2010
UC Santa Cruz Younger Lagoon Reserve <i>Wildlife Intern</i>	June – August 2009

Teaching Experience

Graduate Teaching

- UCD, Introduction to Environmental Plants, Fall 2024
- UCD, Environmental Plant Families, Spring 2019, 2020, 2021, 2022, 2023, 2024
- UCD, Woody Plants in the Landscape: Growth, Ecology & Management, Winter 2023
- UCD, Botany and Physiology of Cultivated Plants, Fall 2021, 2022, 2023
- UCD, Environmental Remote Sensing, Winter 2019, 2022
- UCD, Trees of the Urban Forest, Fall 2018, 2019
- UCD, Introduction to Geographic Information Systems, Fall 2017

Guest Lecturer

- Invited speaker, Stanford Continuing Studies course "Solving Climate Change: Promising Solutions of the 21st Century", Fire ecology and California. April 2021

- Invited speaker, Stanford Continuing Studies course “Solving the Climate Crisis: Action Plans for Sustainable Change”, Fire in California: what’s going on and what can we do? February 2021
- Science lecturer, California College of the Arts “Nature as Literary Inspiration” Fall 2013

Training

Geo for Good 2023 , Google, attendee	October 2023
Cal-TREX Bay Area , Fire Forward/Bay Area Prescribed Fire Council	October - December 2020
Basic Wildland Firefighter 2 (NWCG FFT2)	October 2020
CPR, epinephrine certification	May 2019
Wilderness First Aid, NOLS	June 2018

Professional Affiliations

Association for Fire Ecology	December 2023 - 2024
Ecological Society of America	January 2020 - 2022
American Geophysical Union	August 2019 - 2022
UC Davis Data Science Initiative	October 2018 – September 2020
<i>Graduate Student Affiliate</i>	